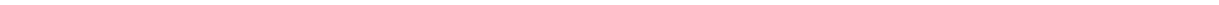


University of Heidelberg
Faculty of Physics and Astronomy
Beginners Physics Lab 1

Experiment 13 – Resonance

Author:	Jonathan Gießen
Group:	Group 8
Supervisor:	Fabian Mrosek
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AI-Disclaimer

AI tools were used to assist in translating, as well as in the generation of text in section 1

1. Introduction

In this experiment, the fundamental dynamic properties of oscillating systems are investigated using a torsion pendulum. The focus is on both the free, damped oscillation and the forced oscillation under the influence of a periodic external torque.

For the experimental setup, the torsion pendulum is driven by a stepper motor with an eccentric, while an eddy current brake (based on Lenz's law) provides a controllable damping that is proportional to the velocity. The aim of the experiment is to determine the oscillation period and the damping constants of the system. Furthermore, the resonance behavior during forced oscillation is analyzed to experimentally determine characteristic quantities such as the resonance frequency, the full width at half maximum (FWHM) of the resonance curve, and the resonance amplitude enhancement, and to compare them with theoretical expectations.

2. Fundamentals

2.1. Free Damped Oscillation

If the torsion pendulum is displaced once and left to itself, it performs a free oscillation. Since frictional forces are present in every real system—which in this experiment (due to the eddy current brake) are proportional to the instantaneous velocity—the amplitude decreases as time progresses. The time dependence of this damped oscillation is described by the following equation:

$$a(t) = a_0 e^{-\delta t} \sin(\omega_f t) \quad (2.1)$$

Here, a_0 denotes the initial amplitude, δ the damping constant, and ω_f the angular frequency of the damped oscillation. The term $e^{-\delta t}$ describes the exponential decay of the oscillation. If one considers only the turning points of the oscillation (where the sine takes the value of one), the temporal decrease of the amplitude is given by:

$$a(t) = a_0 e^{-\delta t} \quad (2.2)$$

The damping constant δ can be determined experimentally by measuring the time $t_{1/2}$ after which the amplitude has dropped to exactly half of its original value. From equation (2) it follows:

$$\delta = \frac{\ln(2)}{t_{1/2}} \quad (2.3)$$

Due to damping, the system oscillates slightly slower than in the completely undamped case. The relationship between the angular frequency of the damped oscillation ω_f and the natural angular frequency of the undamped oscillation ω_0 is:

$$\omega_f^2 = \omega_0^2 - \delta^2 \quad (2.4)$$

2.2. Forced Oscillation and Resonance

If a periodic torque with the driving frequency ω is imposed on the system via a mechanical transmission system, it is referred to as a forced oscillation. After the initial transient process has decayed, the pendulum enters a stationary oscillation whose frequency exactly matches the driving frequency ω .

The amplitude $b(\omega)$ of this forced oscillation strongly depends on the applied frequency and is described by the resonance curve:

$$b(\omega) = \frac{A\omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (2\delta\omega)^2}} \quad (2.5)$$

Here, A represents the constant amplitude of the exciter (where for the static case $\omega \rightarrow 0$, the approximation $\omega_0 \approx \omega_f$ can be assumed). By differentiating this function, the resonance frequency ω' at which the pendulum's amplitude reaches its maximum can be determined:

$$\omega'^2 = \omega_0^2 - 2\delta^2 \quad (2.6)$$

As can be seen from equation (6), due to the damping, the amplitude maximum ω' occurs at a slightly lower frequency than the undamped natural frequency ω_0 .

Two other central quantities characterize the shape of the resonance curve:

- **Full Width at Half Maximum (FWHM) H :** It describes the frequency width of the resonance curve at the height $b(\omega')/\sqrt{2}$. For systems that are not too strongly damped, the following approximation applies:

$$H = 2\delta \quad (2.7)$$

- **Resonance Step-Up (Quality Factor):** It is defined as the ratio of the maximum amplitude at resonance to the static amplitude at a frequency of zero ($\omega \rightarrow 0$):

$$\frac{b(\omega')}{b(\omega \rightarrow 0)} = \frac{\omega_0}{2\delta} \quad (2.8)$$

2.3. Transient Process

The complete mathematical solution of the equation of motion for a harmonic excitation results from the superposition of the free, decaying oscillation and the stationary, forced oscillation:

$$\alpha(t) = a_0 \sin(\omega_f t - \beta) e^{-\delta t} + b(\omega) \sin(\omega t - \epsilon) \quad (2.9)$$

The first term of this equation describes the transient process. The parameters β (depending on the initial state) and ϵ (the phase difference between the exciter and the system) determine the exact phase. As soon as the damped component has practically decayed to zero ($e^{-\delta t} \approx 0$), only the stationary oscillation remains, which is relevant for measuring the resonance curve.

3. Messprotokoll

3.1. Versuchsaufbau und Aufgaben

23.09.2026

Theo Senoner & Jonathan Gießen

V13: Resonance

Tutor: Wiktor Poklepa

T1:

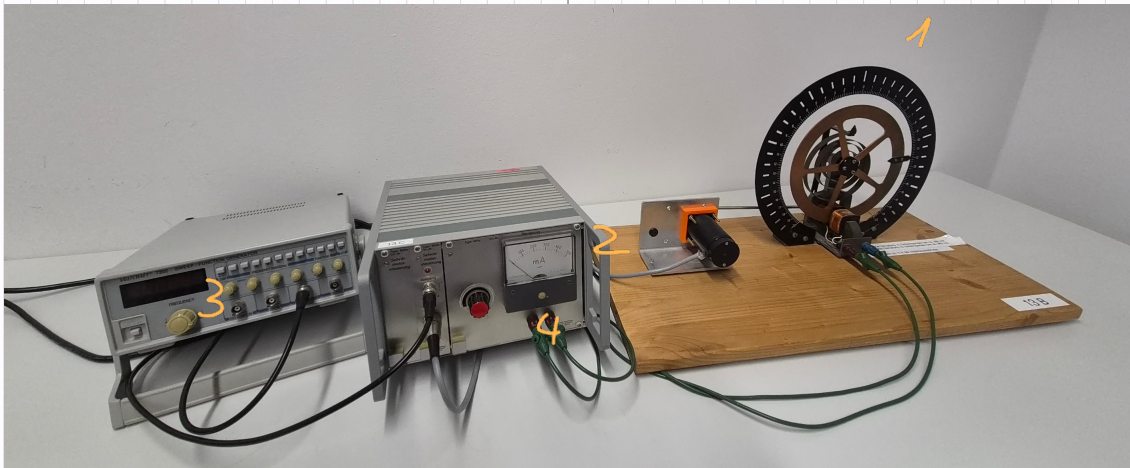


Fig 3.1: Setup

- 1 • Rotary pendulum (Pohl's wheel), stimulated by a stepper motor with an eccentric
- 2 • Stepper motor drive controller with power supply
- 3 • Function generator
- 4 • Power supply unit for the eddy current brake

T2: $\Delta t = \pm 0,25s$ Tab 3.2: Oscillation period

Messung	n	t [s]
1	20	39,61
2	20	39,72
3	20	39,73

T3: $\Delta t = \pm 0,5s$

Tab 3.3 Transient time

Current [mA]	Oscillations until 5% Amplitude [n]	Time [s]
$I_1: 280$	14	27,7
$I_2: 360$	9	17,8
$I_1: 280$	14	26,0
$I_2: 360$	9	17,7

Δt is higher because reading off the amplitude comes with uncertainty, too

$\Delta I = \pm 12,5 \text{ mA}$ (b. the scale)

T4: $\Delta L = 0.2 \text{ cm}$ Tab. 3.4: Amplitude decrease (damped)

n	Ampl. [cm] I_1	Ampl. [cm] I_2	Ampl. [cm] I_1	Ampl. [cm] I_2
0	20,0	20,0	20,0	20,0
1	16,5	16,2	14,6	14,8
2	13,6	13,5	10,7	10,7
3	11,2	11,1	7,8	7,8
4	9,2	9,0	5,6	5,6
5	7,6	7,4	4,1	4,1
6	6,2	6,0	2,9	2,9
7	5,0	5,0	2,0	2,0
8	4,0	4,0	1,4	1,4
9	3,2	3,2	1,0	1,0
10	2,6	2,6	0,7	0,6
11	2,1	2,0	0,5	0,2
12	1,7	1,6	0,3	
13	1,3	1,3		
14	1,0	1,0		
15	0,8	0,8		
<u>16</u>	0,6	0,6		
<u>17</u>	0,5	0,4		

T5:

 $\Delta l = \pm 0,2 \text{ cm}$ Tab 3.5: Amplitude decrease (forced) I_1

$f [\text{Hz}]$	Ampl. $b [\text{cm}] (I_1)$	$f [\text{Hz}]$	Ampl. $b [\text{cm}] (I_1)$
300	0,4	2150	2,6
500	0,4	2200	2,0
700	0,4	2250	1,6
900	0,5	2300	1,3
1100	0,55	2500	0,8
1300	0,7	2700	0,6
1500	0,9	2900	0,4
1600	1,0	3100	0,3
1700	1,3		
1800	1,8		
1850	2,2		
1900	2,8		
1950	3,8		
1960	4,3		
1970	4,6		
1980	4,9		
1990	5,3		
2000	5,7		
2010	5,9		
2020	6,1		
2030	6,1		
2040	6,0		
2050	5,7		
2060	5,2		
2070	4,8		
2080	4,5		
2090	4,2		
2100	3,8		

Tab 3.6: Amplitude decrease (forced) I_2

f [Hz]	Ampl. b [cm] (I_2)	f [Hz]	Ampl. b [cm] (I_2)
300	0,4	2150	2,3
500	0,5	2200	1,8
700	0,5	2250	1,5
900	0,55	2300	1,3
1100	0,6	2500	0,7
1300	0,7	2700	0,6
1500	0,8	2900	0,4
1600	1,0	3100	0,4
1700	1,3		
1800	1,7		
1850	2,0		
1900	2,5		
1950	3,1		
1960	3,2		
1970	3,3		
1980	3,4		
1990	3,6		
2000	3,7		
2010	3,8		
2020	3,8		
2030	3,7		
2040	3,7		
2050	3,6		
2060	3,5		
2070	3,4		
2080	3,2		
2090	3,0		
2100	2,9		

2. RW

3.2. Procedure

The measuring process is divided into four main tasks to analyze both free and forced oscillations:

1. **Period of the Free Oscillator:** Determine the oscillation period T_0 of the undamped, free torsion pendulum by measuring the duration of 20 full oscillation periods. Repeat this measurement three times.
2. **Transient Time and Damping:** Activate the eddy current brake. Find two specific damping currents: one where the amplitude decays to 5% of its initial value after about 10 oscillations, and another where it decays to 5% after about 15 oscillations. Measure the transient time for both settings.
3. **Amplitude Decay:** For the two selected damping currents, release the pendulum from a reversal point at $t = 0$ and record the amplitude after every full period to document the exponential decay.
4. **Forced Oscillations:** Excite the pendulum with the stepper motor. Measure the stationary amplitude as a function of the excitation frequency (generator frequency from 300 Hz to 4000 Hz) for both damping settings. Wait for the transient process to finish before each reading, and use smaller frequency steps (50 Hz) near the resonance peak to accurately determine the resonance frequency and the Full Width at Half Maximum (FWHM).

4. Auswertung

4.1. Excercise 1: Period of the Free Oscillator

We measured the oscillation period of the undamped torsion pendulum by timing 20 full oscillations. The measurements were repeated three times to ensure accuracy. The results are summarized in Table 4.1. The measured Values are:

Table 4.1: Measured oscillation periods for 20 full oscillations of the undamped torsion pendulum.

Measurement	Time for 20 Oscillations (s)	Calculated Period T_0 (s)
1	39.61 ± 0.25	1.981(13)
2	39.72 ± 0.25	1.986(13)
3	39.73 ± 0.25	1.987(13)

With the error calculation of:

$$\Delta T_0 = \sqrt{\Delta T_{\text{stat}}^2 + \Delta T_{\text{sys}}^2} = \sqrt{(0.002)^2 + (0.013)^2} \approx 0.014 \text{ s} \quad (4.1)$$

The Mean of the measured periods is:

$$T_0 = 1.98413 \quad (4.2)$$

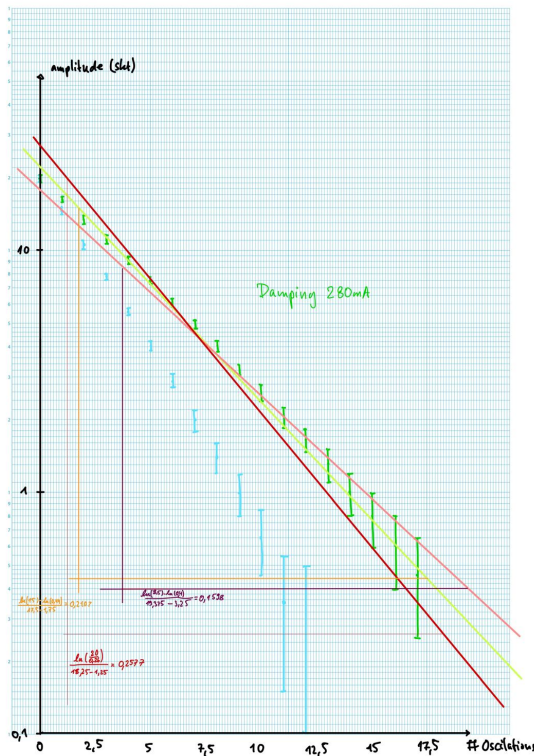
Result

The oscillation period of the undamped torsion pendulum is determined to be:

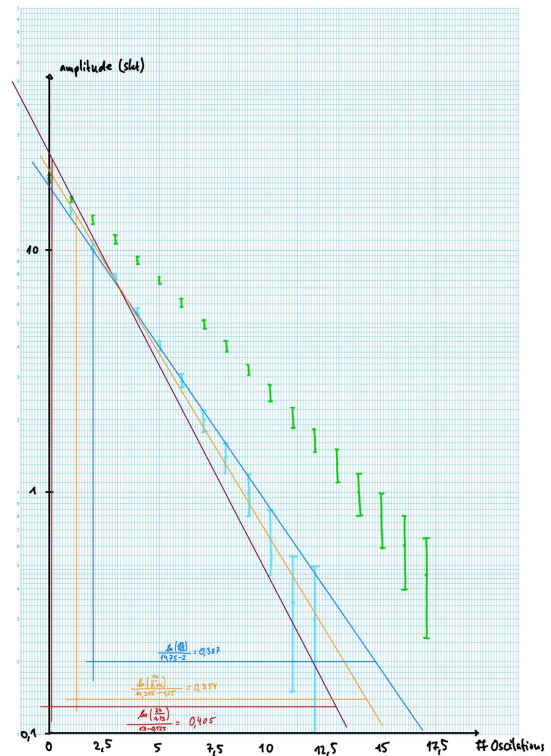
$$T_0 = (1.984 \pm 0.014) \text{ s} \quad (4.3)$$

4.2. Exercise 2: Evaluate the damping constant δ

The damping constant δ can be determined by measuring the time $t_{1/2}$ after which the amplitude has decreased to only 5% of its initial value. For each damping current of the eddy current brake, the average of two measurements of the amplitude was determined after each full oscillation until it reached 5% of the initial amplitude. The results are plotted logarithmically to determine the damping constant δ from the slope of the fit. The error of the amplitude is estimated to be $\Delta a = 0.2 \text{ Skt}$, which corresponds to the uncertainty of the measurement due to the scale division of the torsion pendulum.



(a) Damping current: 280 mA.



(b) Damping current: 360 mA.

Figure 4.1: Determination of the damping coefficient for two damping currents.

¹The plots are listed in full scale in the appendix

Result

The damping constants for the two damping currents are determined to be:

$$\delta_{280} = (-0.21 \pm 0.05) \frac{1}{\text{s}} \quad (4.4)$$

$$\delta_{360} = (-0.35 \pm 0.05) \frac{1}{\text{s}} \quad (4.5)$$

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References

- [1] Dr. J.Wagner, *Physikalisches Anfängerpraktikum*, Stand 11/2024
- [2] W. Ilberg, M. Krötzsch, *Physikalisches Praktikum für Anfänger*, BSB B. G. Teubner Verlagsgesellschaft, Leipzig (1985).

A. Plots

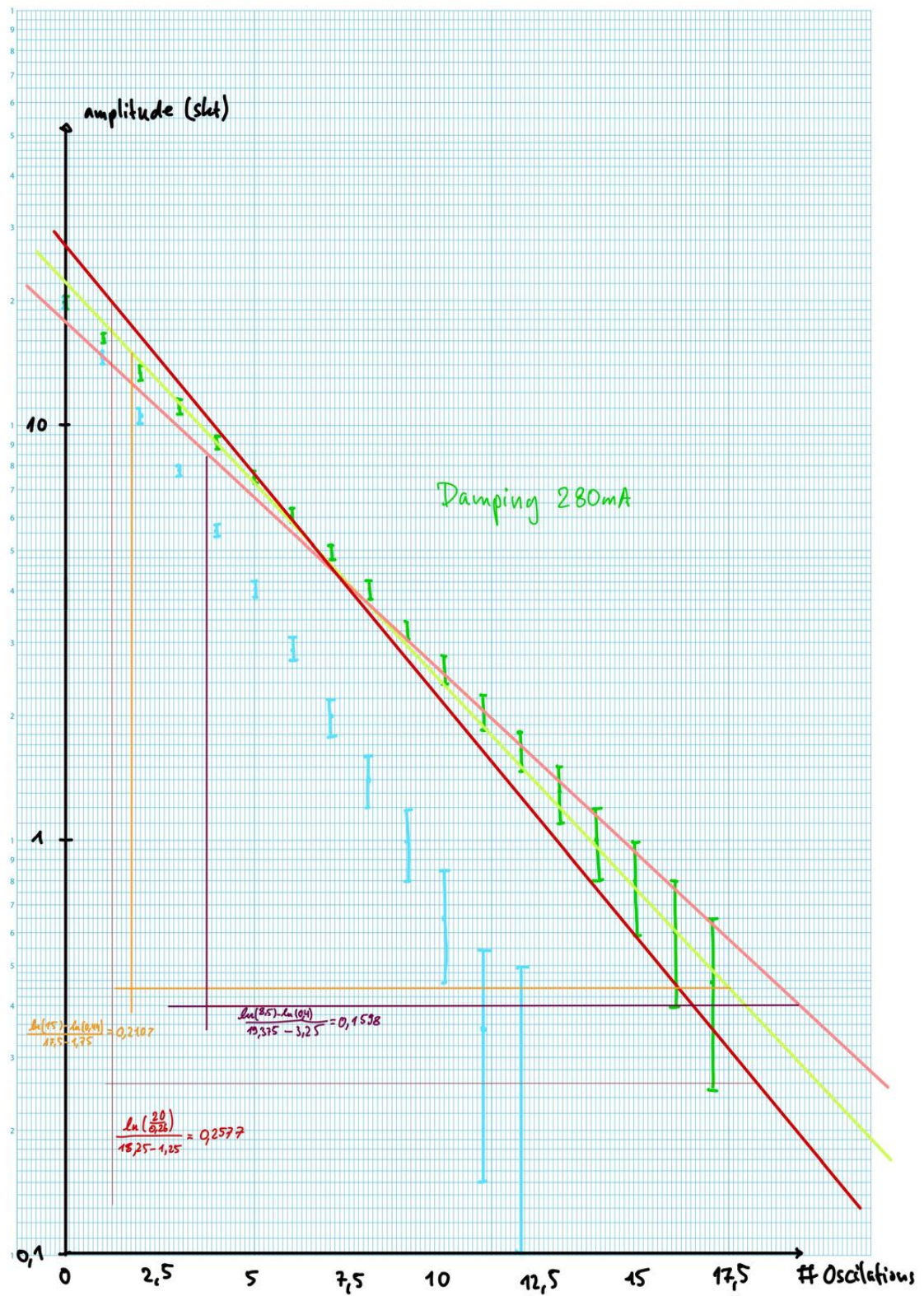


Figure A.1: Amplitude decay for a damping current of 280 mA.



Figure A.2: Amplitude decay for a damping current of 360 mA.